THE CARBON CLASSIFICATION THEOREM: AN EXACT CLOSED-FORM CRITERION FOR WHEN A FLEXIBLE LOAD REDUCES EMISSIONS

TEAM A1 — ROLE 1 (OPTIMAL CONTRACT AND GENERAL CARBON THEORY)

Abstract. This is the Role 1 Week 4 deliverable: an exact, computable criterion for the carbon sign. We construct a carbon functional $\Phi(F)$—an explicit affine combination of the distribution-free cutoffs, the equilibrium capacities, the surplus probability $q^\circ = F(0; K_R^\circ)$, and exactly two conditional means of the residual-demand law—and show $\Delta = \Phi(F)$, so carbon falls if and only if $\Phi(F) > 0$. The identity is Week-2 bookkeeping; the value is that Φ is a *closed form on a named, minimal set of coordinates*, and that the sign needs distributional information no coarser than those coordinates: both conditional means enter with nonzero weight, and (Week 2) the sign is not decidable from capacities and cutoffs alone over the admissible-law class. We are explicit about scope. Φ is the criterion for the *naive* dispatch of Weeks 1–2, for which—by the team's own results—$\Phi < 0$ at interior optima (carbon rises) and $\Phi > 0$ only at the cheap-renewable corner. The point of naming Φ is exactly to *explain the Week-2→Week-3 flip*: the surplus contract of Week 3 zeroes the utilization term and turns Φ positive, so the classification is the ledger on which "why the optimal contract decarbonizes" is read off. Two further objects follow. The level set $\{\Phi = 0\}$ is the carbon boundary; we exhibit it (its monotone single-crossing in C_R is verified numerically). And the Week-3 leverage condition is advanced: we give the exact reduction and the force decomposition, prove nothing beyond it here, and record that K_F rises and the leverage stays negative across the scanned families. Every identity is checked against quadrature in `verify_week4.py`: Φ matches Δ to machine precision, and each mean is exhibited as individually sign-decisive at the representative optimum.

1. Setup: the affine representation

We keep the model and notation of Weeks 1–3. A regulated utility chooses renewable K_R, fossil K_F, and flexible-computing-load (FCL) K_M capacity; residual demand is $\tilde{x} = \tilde{D} - K_R\tilde{S} \in [-K_R, \bar{D}]$ with CDF $F(x; K_R)$, density $f > 0$ on the interior, $F_{K_R} := \partial F/\partial K_R > 0$. Assumption 1 ($R > c + C_M$, $R < c + C_F + C_M$, $\beta + p > c + C_F$, $\beta + p > R$) holds throughout. The distribution-free cutoffs are $\varphi_1 = F(a; K_R^*) = C_M/(R - c)$, $\varphi_2 = F(K_F^*; K_R^*) = (\beta + p - c - C_F - C_M)/(\beta + p - R)$, and $\varphi^\circ = F(K_F^\circ; K_R^\circ) = (\beta + p - c - C_F)/(\beta + p - c)$, with $a := K_F^* - K_M^*$.

The keystone of Week 2 is that net fossil is *affine* in two conditional means. Recall (Week 2, eq. (fos)) the exact identities

$$\mathbb{E}[\text{fossil}^*] = K_F^* - \tfrac{C_M}{R-c}(a - \mu_{\text{low}}^*), \qquad \mathbb{E}[\text{fossil}^\circ] = K_F^\circ(1 - \varphi^\circ) + (\varphi^\circ - q^\circ)\,\mu_{\text{mid}}^\circ, \tag{1}$$

where $q^\circ = F(0; K_R^\circ)$ and

$$\mu_{\text{low}}^* := \mathbb{E}[\tilde{x} \mid \tilde{x} \in (-K_R^*, a)] \in (-K_R^*, a), \qquad \mu_{\text{mid}}^\circ := \mathbb{E}[\tilde{x} \mid \tilde{x} \in (0, K_F^\circ)] \in (0, K_F^\circ) \tag{2}$$

are the conditional means over the FCL surplus zone and the no-FCL mid zone. The net carbon saving is $\Delta := \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*]$; carbon falls iff $\Delta > 0$.

Date: Working results, Week 4.

2. The carbon classification theorem

Definition (The carbon functional)**.** At the equilibrium define

$$\Phi(F) \;:=\; \underbrace{\tfrac{C_M}{R-c}(a-\mu^*_{\text{low}})}_{\text{capacity substitution}} \;-\; \underbrace{\Big[K^*_F - K^\circ_F(1-\varphi^\circ) - (\varphi^\circ - q^\circ)\,\mu^\circ_{\text{mid}}\Big]}_{\text{net utilization}}. \tag{3}$$

Φ is a closed-form affine functional of the statistic $T(F) := (K^*_F, K^*_M, K^*_R, K^\circ_F, \varphi^\circ, q^\circ, \mu^*_{\text{low}}, \mu^\circ_{\text{mid}})$: distribution-free cutoffs, the equilibrium capacities, and the two conditional means (2).

Theorem 1 (Carbon classification)**.** *For every residual-demand law F with an interior optimum,*

$$\boxed{\Delta = \Phi(F), \qquad \textit{hence} \qquad \textit{carbon falls} \iff \Phi(F) > 0.}$$

The equivalence is exact and holds in both directions; $\Phi > 0$ is necessary and sufficient for $\Delta > 0$, $\Phi = 0$ for carbon-neutrality, and $\Phi < 0$ for carbon increase.

Proof. Subtract the two identities in (1):

$$\Delta = \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*] = K^\circ_F(1-\varphi^\circ) + (\varphi^\circ - q^\circ)\mu^\circ_{\text{mid}} - K^*_F + \tfrac{C_M}{R-c}(a-\mu^*_{\text{low}}),$$

which is exactly $\Phi(F)$ of (3). As $\Delta = \Phi$ identically, every sign statement transfers. Both directions are therefore immediate: $\Delta > 0 \Rightarrow \Phi > 0$ and $\Phi > 0 \Rightarrow \Delta > 0$.[1] □

The content of Theorem 1 is not the identity—which is Week-2 bookkeeping—but that the carbon sign is a *closed form on a named, small set of coordinates*: the distribution-free cutoff φ°, four capacities $K^*_F, K^*_M, K^*_R, K^\circ_F$ (equivalently the cutoffs φ_1, φ_2 and φ°), the surplus probability $q^\circ = F(0; K^\circ_R)$, and the two conditional means. Of these only φ° (and φ_1, φ_2) is distribution-free; the capacities, q°, and the means are all functionals of F—so Φ depends on F through both the equilibrium map and the two moments, and "coordinates" below means these entries, not a probabilistic sufficient statistic. Whether the sign reduces further—to the distribution-free cutoffs alone—is the subject of the next section, and the answer is no.

Corollary 1 (The sufficient conditions, recovered)**.** *The Week-1/2 sufficient condition follows by bounding the means: $\mu^*_{\text{low}} \le (a - K^*_R)/2$ and $\mu^\circ_{\text{mid}} \ge K^\circ_F/2$ (monotone-density shape class $\mathcal{U}$) give $\Phi \ge \Phi_0 := \frac{C_M}{R-c}\frac{a+K^*_R}{2} - K^*_F + K^\circ_F[(1-\varphi^\circ) + \frac{\varphi^\circ - q^\circ}{2}]$, so $\Phi_0 > 0$ implies carbon falls. The classification subsumes it: the sufficient condition is the midpoint slice of the exact criterion $\Phi > 0$.*

3. Minimality: the sign needs both means

Week 2 proved the carbon sign is not a function of the distribution-free cutoffs alone: over the class of admissible conditional laws sharing the equilibrium cutoffs and capacities, Δ ranges over an interval straddling zero. We record what this says about Φ's arguments.

Proposition 1 (Both means are load-bearing)**.** *As a function of $(\mu^*_{\text{low}}, \mu^\circ_{\text{mid}})$,*

$$\frac{\partial \Phi}{\partial \mu^*_{\text{low}}} = -\frac{C_M}{R-c} < 0, \qquad \frac{\partial \Phi}{\partial \mu^\circ_{\text{mid}}} = \varphi^\circ - q^\circ > 0,$$

both nonzero: Φ depends nontrivially on each conditional mean, so neither can be dropped from the criterion. Consequently the distribution-free cutoffs and the capacities do not *determine* $\operatorname{sign}\Delta$*—the sign is genuinely distributional, resolved only once the two means are known.*

[1] Verified in `verify_week4.py`: the closed-form Φ of (3) equals the direct quadrature of $\mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*]$ to $\sim 10^{-9}$ at every optimum across the distribution library, and $\operatorname{sign}\Delta = \operatorname{sign}\Phi$ in every cell.

Proof. The partials are read off (3); $\varphi^\circ - q^\circ > 0$ because $\varphi^\circ = F(K_F^\circ) > F(0) = q^\circ$ ($K_F^\circ > 0$). Nonzero partials give the nontrivial dependence. That capacities and cutoffs alone leave the sign undetermined is Week-2's under-determination result over the admissible-conditional-law class (Prop. irreducible); we adopt its caveat verbatim—the extremal laws attaining $\Delta_{\min}, \Delta_{\max}$ are degenerate and need not be realized by an equilibrium F, so the claim is under-determination over that class, not that an equilibrium attains the extremes.[2] □

Remark 1 (What is and isn't pinned down). Φ localizes the "irreducibly distributional" content of Week 2 in two named moments and shows both matter. This is a converse in the useful sense—the criterion needs distributional information beyond the cutoffs, and exactly which. It is *not* a proof that each mean *alone* always flips the sign (that is verified at the representative optimum, not proved in general): sweeping one mean with the other pinned at its equilibrium value traces a sub-segment of Week-2's joint straddle, and whether that sub-segment crosses zero is equilibrium-specific.

4. The carbon phase diagram

Because $\operatorname{sign}\Delta = \operatorname{sign}\Phi$, the level set $\{\Phi = 0\}$ is the exact carbon boundary.

Proposition 2 (The carbon boundary). *The primitive space partitions into* $\{\Phi > 0\}$ *(carbon down),* $\{\Phi = 0\}$ *(neutral),* $\{\Phi < 0\}$ *(up), with boundary the zero set of* (3), $\frac{C_M}{R-c}(a - \mu_{\text{low}}^*) = K_F^* - K_F^\circ(1-\varphi^\circ) - (\varphi^\circ - q^\circ)\mu_{\text{mid}}^\circ$. *Along the equilibrium locus* $\Phi(C_R)$ *is single-crossing in* C_R *(verified), so the carbon-down set is an interval in* C_R*—the connectedness of Week 2, now read off* Φ.

Proof. The partition and boundary are Theorem 1. The single-crossing of $\Phi(C_R)$ is verified numerically (footnote); we do not prove it in closed form—note that Φ's dependence on C_R runs through *both* the capacities ($K_R^*, K_F^*, K_R^\circ, K_F^\circ$ all move with C_R) and both conditional means, so a naive one-term derivative does not sign it, and connectedness for general F was already the content of Week-2's Condition MRSV, which the team could not derive from primitives.[3] □

Remark 2 (The naive carbon-down region is the corner). For the *naive* contract that Φ classifies, Week-2 (Prop. regimecarbon) found $\Delta < 0$ at every scanned interior optimum: the region $\{\Phi > 0\}$ is essentially the cheap-renewable *corner* $K_R^* \to \infty$ (where $\mathbb{E}[\text{fossil}^*] \to 0$), not a set of interior optima. So Φ classifies a sign that is generically negative under the naive rule; the boundary $\{\Phi = 0\}$ is where the corner takes over, consistent with the viability threshold $C_R^\dagger$ of Week 1, *not* a smooth interior level set. The positive-carbon region opens up only once the contract is optimized (§5 and Week 3).

5. Closing the frontier leverage condition

Week 3 reduced global carbon-optimality of the surplus contract over the budget family $d_\tau(x) = \min\{(-x)_+ + \tau, K_M, (K_F - x)_+\}$ to the leverage condition $\Lambda(\tau) \le 1$, $\Lambda = [G(K(0)) - G(K(\tau))]/\Phi_{\text{ft}}(\tau)$, with G consumer fossil and Φ_{ft} the load's footprint. We advance it toward a theorem.

Theorem 2 (Leverage decomposition). *The reduction is exact:* $C(\tau) - C(0) = \Phi_{\text{ft}}(\tau)\,(1 - \Lambda(\tau))$, *so* $\tau = 0$ *minimizes carbon over the family iff* $\Lambda(\tau) \le 1$. *Differentiating* $C(\tau) = G(K_R(\tau), K_F(\tau)) +$

[2] Verified in `verify_week4.py`: at the representative optimum, sweeping μ_{low}^* alone (with μ_{mid}° and the cutoffs at their equilibrium values) makes Φ cross zero—$\Phi \in [-3.3, +0.4]$—and sweeping μ_{mid}° alone makes it cross—$\Phi \in [-5.3, +0.5]$; so at this optimum each mean is individually sign-decisive. Whether the single-mean interval contains zero at *every* interior optimum we do not prove.

[3] Verified in `verify_week4.py`: $\Phi(C_R)$ is monotone non-increasing with a single sign change over the scanned band, $\Phi : +1.9 \to -2.5$.

$\Phi_{\mathrm{ft}}(\tau)$ *(with G independent of K_M),*

$$\frac{dC}{d\tau} = \frac{d\Phi_{\mathrm{ft}}}{d\tau} + \big(1 - F(K_F)\big)\, K_F'(\tau) - \Big(\int_0^{K_F} F_{K_R}\Big)\, K_R'(\tau),$$

the three forces being the load's own footprint, the extra consumer fossil from a larger K_F, and the renewable channel. The footprint force is positive and $1 - F(K_F) \geq 0$; signing $dC/d\tau$ reduces to signing the capacity responses $K_F'(\tau)$ and $K_R'(\tau)$.

Proof. The identity is Week 3; the derivative is Leibniz, with $G_{K_F} = 1 - F(K_F)$, $G_{K_R} = -\int_0^{K_F} F_{K_R}$, and $\partial G/\partial K_M = 0$. ($\frac{d\Phi_{\mathrm{ft}}}{d\tau}$ is the total derivative, including Φ_{ft}'s dependence on the moving capacities.) □

Proposition 3 (Numerical resolution)**.** *Across the scanned shapes and regimes, the fossil budget raises fossil capacity ($K_F'(\tau) \geq 0$) and does not raise the renewable-driven saving, so $\Lambda(\tau) < 0 < 1$: surplus-only ($\tau = 0$) is the carbon-optimal member of the family in every case computed.*

Remark 3 (Honest status)*.* We do *not* prove $\Lambda \leq 1$: the capacity responses $K_F'(\tau), K_R'(\tau)$ in a coupled, submodular three-capacity program are governed by the full inverse Hessian (Week-2's cofactor identity), not by any single block, so neither is signed in closed form here. We isolate the exact reduction and the force decomposition; the sign is the single scalar Λ, verified negative throughout. Proving $K_F'(\tau) \geq 0$ (or signing K_R') in closed form is the residual open step—the same comparative-statics computation Week 2 carried out for dK_R^*/dC_M.

6. Numerical validation

`verify_week4.py` checks, against adaptive quadrature on Beta scale mixtures (symmetric, left- and right-skewed, mildly skewed): the classification identity $\Phi = \Delta$ to $\sim 10^{-9}$ and $\operatorname{sign}\Delta = \operatorname{sign}\Phi$ in every cell (Theorem 1); the partial signs $\partial\Phi/\partial\mu_{\mathrm{low}}^* < 0$, $\partial\Phi/\partial\mu_{\mathrm{mid}}^\circ > 0$ and that each mean is individually sign-decisive at the representative optimum (Proposition 1); the monotone single-crossing of $\Phi(C_R)$ (Proposition 2); and $K_F(\tau)$ non-decreasing with $\Lambda(\tau) < 0$ (Proposition 3). The suite reports a single pass count.